Introduction to Decision Support System (DSS)

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Management Support Systems An Overview

Emerging and Advanced Computer Technologies for Supporting Managerial Problem Solution

- **Changing Organizational Structure**
- **Enabling Business Transformation**
- **Changing Management Methods**

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Managers and Decision Making: Why Computerized Support?

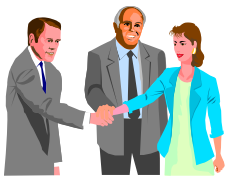
- **Competition**
- **Speed**
- **The *MANAGERS* are <u>always</u> responsible for decision making**

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The Nature of Managers' Work

[Make Decisions!] Mintzberg (1980) (Table 1.1) Roles

- **Interpersonal**
  - **Figurehead**
  - **Leader**
  - **Liason**
- **Informational**
  - **Monitor**
  - **Disseminator**
  - **Spokesperson**
- **Decisional**
  - **Entrepreneur**
  - **Disturbance Handler**
  - **Resource Allocator**
  - **Negotiator**



Managers *need* information and *use* computers to support decision making

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Managerial Decision Making and Information Systems

- ***Management*** **is a process by which organizational goals are achieved through the use of resources**



- **Resources:** ***Inputs***
- **Goal Attainment:** ***Output***
- **Measuring Success:**

Productivity = Outputs / Inputs

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Factors Affecting Decision Making (Figure 1.1)

- **Technology / Information / Computers**
- **Structual Complexity / Competition**
- **International Markets / Political Stability / Consumerism**
- **Changes, Fluctuations**

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Managers and Computerized Support

- ***Information Technology:*** **vital to the business**
- **Support technologies extensively implemented**

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Need for Computerized Decision Support and the Supporting Technologies

- **Speedy computations**
- **Overcome cognitive limits in processing and storage**
- **Cognitive limits may restrict an individual's problem-solving capability**
- **Cost reduction**
- **Technical support**
- **Quality support**
- **Competitive edge**

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Decision Support Technologies

Management Support Systems (MSS)

- Decision Support Systems (DSS)
- Group Support Systems (GSS)
- Enterprise (Executive) Information Systems (EIS)
- Enterprise Resource Planning (ERP) and Supply-Chain Management (SCM)
- Knowledge Management Systems
- Expert Systems (ES)
- Artificial Neural Networks (ANN)
- Hybrid Support Systems
- Intelligent DSS

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Framework for Decision Support

- Figure 1.2 (Gorry and Scott Morton, 1971)

Combination of

- Simon (1977) Taxonomy
- Anthony (1965) Taxonomy

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Decision Support Framework

Type of Decision	Type of Control: Operational Control	Managerial Control	Strategic Control
Structured			
Semistructured			
Unstructured			

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Gorry & Scott-Morton Framework of DSS

FIGURE 1.2 Decision Support Frameworks

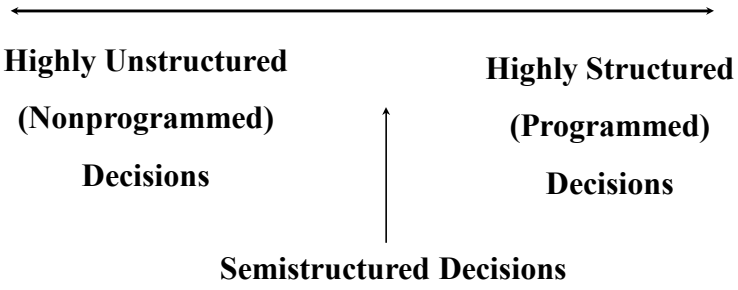
Type of Decision	Type of Control: Operational Control	Managerial Control	Strategic Planning
Structured	1 Accounts receivable, accounts payable, order entry	2 Budget analysis, short-term forecasting, personnel reports, make-or-buy	3 Financial management (investment), warehouse location, distribution systems
Semistructured	4 Production scheduling, inventory control	5 Credit evaluation, budget preparation, plant layout, project scheduling, reward system design, inventory categorization	6 Building new plant, mergers and acquisitions, new product planning, compensation planning, quality assurance planning, HR policies, inventory planning
Unstructured	7 Selecting a cover for a magazine, buying software, approving loans, help desk	8 Negotiating, recruiting an executive, buying hardware, lobbying	9 R & D planning, new technology development, social responsibility planning

Which of the cells has the most computer systems in use today? Why?

Any more example decisions for cells 1-9?

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Decision Making Along a Continuum (Simon)



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Three Phase Decision-making Process (Simon)

- Intelligence--searching for conditions that call for decisions
- Design--inventing, developing, and analyzing possible courses of action
- Choice--selecting a course of action from those available

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- *Unstructured problem* has no structured phases
- *Semistructured problem* has some (or some parts with) structured phases
- *Structured problem* has *all* structured phases
  - Procedures for obtaining the best solution are known
  - Objectives are clearly defined
  - Management support systems can be useful

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- *Unstructured problems* often solved with human intuition
- *Semistructured problems* in between

  Solve with standard solution procedures and human judgment
- A Decision Support System can help managers understand problems in addition to providing solutions
- *Goal of DSS:* Increase the effectiveness of decision making

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Anthony's Taxonomy (1965)

- Encompass *ALL* managerial activities
  - Strategic planning
  - Management control
  - Operational control
- Combine Anthony's and Simon's Taxonomies
- DSS for semistructured and unstructured decisions
- MIS and management science approaches insufficient

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Computer Support for Structured Decisions



- Since the 1960s
- Repetitive in nature
- High level of structure
- Can abstract and analyze them, and classify them into prototypes
- Solve with quantitative formulas or models
- Management Science (MS) / Operations Research (OR)

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Management Science

***Scientific approach* to automate managerial decision making**

1. **Define *problem***
2. **Classify *problem***
3. **Construct mathematical *model***
4. **Find and evaluate potential solutions**
5. **Choose and recommend a solution**

***Modeling*:** Transforming the real-world problem into an appropriate prototype structure

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Decision Support Systems

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Decision Support Systems Concept

Decisions, decisions, ...

- When do you know, you have to make a decision?
- How will you go about making the decision?
- Why is that intuition and trial&error approaches to managerial decisions may not be effective in today's business environment?
- What are some of the tools that can be helpful in making a decision?

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Decision Support Systems Concept

No universally accepted definition in literature

- "DSS couples the intellectual resources of individuals with the capabilities of the computer to improve the quality of decisions." – Keen
- "DSS are interactive computer-based systems that help decision makers utilize data and models to solve unstructured problems." – Sprague & Carlson
- "An interactive, flexible, and adaptable computer-based information system, especially developed for supporting the solution of a non-structured management problem for improved decision making. It utilizes data, provides an easy-to-use interface, and allows for the decision maker's own insights." - Turban

Thus, DSS is a multi-disciplinary topic covering:

Database research, AI, human-computer interaction, management science modeling, software engineering and telecommunications!

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Decision Support Systems Concept

- **DSS are interactive computer-based systems, which help decision makers utilize *data* and *models* to solve unstructured problems (Scott Morton, 1971).**
- **Decision support systems couple the intellectual resources of individuals with the capabilities of the computer to improve the quality of decisions. It is a computer-based support system for management decision makers who deal with semi-structured problems (Keen and Scott Morton, 1978).**

Content-free expression

- ***There is no universally accepted definition of DSS***

Umbrella term vs. narrow definition (specific technology)

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Major DSS Characteristics

(DSS In Action 1.5: Houston Minerals Case)

- **Initial risk analysis (management science)**
- **Model scrutiny using experience, judgment, and intuition**
- **Initial model mathematically correct, but incomplete**
- **DSS provided very quick analysis**
- **DSS: flexible and responsive. Allows *managerial intuition* and *judgment***

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Why Use DSS?

Perceived benefits

- decision quality
- improved communication
- cost reduction
- increased productivity
- time savings
- improved customer and employee satisfaction

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Major Reasons

- Unstable economy
- Difficulty in tracking numerous business objectives
- Increased competition
- Electronic commerce
- Existing systems did not support decision making
- IS Department is too busy
- Special analysis
- Need accurate information
- Organizational winner
- New or timely information needed
- Mandated by management
- Cost reductions
- End-user computing

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Selected DSS Applications

Company or Application	Description
Cinergy Corporation	The electric utility developed a DSS to reduce lead time and effort required to make decisions in purchasing coal.
RCA	The company developed a DSS called Industrial Relation Information System (IRIS) to help solve personnel problems and issues.
U.S. Army	It developed a DSS to help recruit, train, and educate enlisted forces. The DSS uses simulation that incorporates what-if features.
National Audubon Society	It developed a DSS called Energy Plan (EPLAN) to analyze the impact of U.S. energy policy on the environment.
Hewlett-Packard	The computer company developed a DSS called Quality Decision Management to help improve the quality of its products and services.
Virginia	The state of Virginia developed the Transportation Evacuation Decision Support System (TEDSS) to determine the best way to evacuate people in case of a nuclear disaster at its nuclear power plants.

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Comparison of DSSs and MISs

Factor	DSS	MIS
Problem type	A DSS is good at handling unstructured problems that cannot be easily programmed.	An MIS is normally used only with more structured problems.
Users	A DSS supports individuals, small groups, and the entire organization. In the short run, users typically have more control over a DSS.	An MIS supports primarily the organization. In the short run, users have less control over an MIS.
Support	A DSS supports all aspects and phases of decision making; it does not replace the decision maker—people still make the decisions.	This is not true of all MIS systems—some make automatic decisions and replace the decision maker.
Emphasis	A DSS emphasizes actual decisions and decision-making styles.	An MIS usually emphasizes information only.
Approach	A DSS is a direct support system that provides interactive reports on computer screens.	An MIS is typically an indirect support system that uses regularly produced reports.
System	The computer equipment that provides decision support is usually on-line (directly connected to the computer system) and related to real time (providing immediate results). Computer terminals and display screens are examples—these devices can provide immediate information and answers to questions.	An MIS, using printed reports that may be delivered to managers once a week, may not provide immediate results.
Speed	Because a DSS is flexible and can be implemented by users, it usually takes less time to develop and is better able to respond to user requests.	An MIS's response time is usually longer.
Output	DSS reports are usually screen oriented, with the ability to generate reports on a printer.	An MIS, however, typically is oriented toward printed reports and documents.
Development	DSS users are usually more directly involved in its development. User involvement usually means better systems that provide superior support. For all systems, user involvement is the most important factor for the development of a successful system.	An MIS is frequently several years old and often was developed for people who are no longer performing the work supported by the MIS.

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Group Support Systems (GSS)

- Decisions often made by groups
- Supports groupwork, anytime, anyplace

Also called

- Groupware
- Electronic meeting systems
- Collaborative computing

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Executive Information (Support) Systems (EIS, ESS)

- Organizational view
- Information needs of executives / managers
- Customized user seductive interface
- Timely and effective tracking and control
- Drill down
- Filter, compress, and track critical data / information
- Identify problems / opportunities

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EIS

- Mid-1980s - large corporations
- Now global
- Affordable to smaller companies
- Serves managers as *enterprise-wide systems*

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Expert Systems (ES)

- Experts solve complex problems
- Experts have specific knowledge and experience
- Expert systems mimic human experts
- ES performance comparable to or better than experts in a specialized and usually narrow problem area

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Intelligent Agents

- Help automate various tasks
- Increase productivity and quality
- Learn how you work

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Artificial Neural Systems

- Artificial Neural Networks (ANN):
- Mathematical models of the human brain
- ANN learn patterns in data
- ANN can work with partial, incomplete, or inexact information

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Knowledge Management Systems (KMS)

- Capture and reuse knowledge at the organizational level
- Knowledge repository for storage
- Organizational impacts can be dramatic

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ERP and SCM

- Enterprise Resource Planning (Management)
- Supply Chain Management including Customer Resource Management (CRM)
- Enterprise-level cost cutters

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Cutting Edge Intelligent Systems

- **Genetic Algorithms**
  Work in an evolutionary fashion
- **Fuzzy Logic**
  Continuous logic (NOT just True / False)
- **Intelligent Agents**
  In search engines, e-mail, electronic commerce

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Hybrid Support Systems

- **Combines MSS technologies**
- **Use strengths of each**
- **Goal: *successful solution* of the managerial problem**
- **Tools support each other**
- **Tools can add intelligence to traditional MSS**

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Computerized Decision Aids Evolution and Attributes

- Computerized procedures development aids decision making (Table 1.2)
- DSS supports specific questions (Table 1.3)

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Evolutionary View of CBIS

1. Time Sequence

- **mid-1950s** Transaction Processing Systems (TPS)
- **1960s** MIS
- **1970s** Office Automation Systems
  DSS
- **1980s** DSS Expanded
  Commercial applications of expert systems
  Executive Information Systems
- **1990s** Group Support Systems
  Neural Computing
  Integrated, hybrid computer systems

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2. Computer evolved over time

3. Systemic linkages in how each system processes data into information

Relationship among these and other technologies (Figure 1.3)

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Relationship Among Technologies

- Each technology unique
- Technologies interrelated
- Each supports some aspects of managerial decision making
- Ever expanding role of information technology improving management
- Interrelationship and coordination evolving

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Summary

- DSS has many definitions
- Complexity of managerial decision making is increasing
- Computer support for managerial decision making
- Several MSS technologies including hybrids

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